Motor and Freewheel System Design for Your 50 kg Fibre-Optic Cable Drum

Based on your requirements, I'll use an automatic BLDC direct-drive system powered by a 22.2 V battery as the primary design option.

Your system is intended to:

- Pull optical-fibre cable laid on the ground back onto the drum.
- Wind the cable at a drum speed of about 50 RPM.
- Allow the cable to unwind when needed without the motor resisting it.
- Control cable tension to avoid damaging the fibre.
- Operate automatically in both CW and CCW directions.

1. Design Information for Your Project

Parameter	**Your specification**
Rotating mass of drum and cable	50 kg
Drum core diameter	200 mm
Drum width	300 mm
Shaft diameter	20 mm
Shaft support	Two bearings

Battery	22.2 V LiPo (6S)
Drum speed	50 RPM
Ground surface	Soil
Motor	BLDC
Drive	Direct coupling

The 50 kg figure is the rotating mass. The drum does not need to be lifted, so using only 50×9.81 to calculate motor torque would not be correct.

2. Recommended Mechanical Arrangement

I recommend this arrangement:

1. BLDC motor
2. Flexible shaft coupling
3. Drum shaft
4. Two pillow-block bearings
5. Cable drum
6. Guide roller
7. Tension-measuring roller
8. Encoder
9. Freewheel / brake mechanism

Important: The two bearings support the radial load from the drum and cable. The motor coupling should not support the drum’s weight.

3. Motor Selection and Torque Calculation

Pulling cable over soil can require more force because of friction, small stones, or uneven ground.

For a preliminary calculation only, let’s assume a pulling force of 100 N. This is not a measured value.

Assume the drum’s effective radius while winding is 150 mm:

$$ T=F\times r $$$$ T=100\times0.15=15\ \text{Nm} $$

With a design factor of 2:

$$ T_{\text{design}}=15\times2=30\ \text{Nm} $$

So, an initial design could be checked against a torque target of 30 Nm. Do not treat this as a final motor specification.

Preliminary motor specifications

Parameter	Preliminary target
Type	BLDC direct-drive torque motor
Voltage	22.2 V nominal, compatible with 6S
Drum speed	Adjustable, maximum 50 RPM
Torque	Approximately 15 Nm estimated

	running torque
Peak design target	Approximately 30 Nm
Direction	CW and CCW
Gearbox	None
Controller	Bidirectional FOC / VESC-type

At 50 RPM and 15 Nm torque:

$$ P=\frac{2\pi NT}{60} $$$$ P=\frac{2\pi\times50\times15}{60} $$$$ \boxed{P=78.5\ \text{W}} $$

At 30 Nm, the same mechanical power would be approximately 157 W.

This does **not** mean that a 100 W motor will be sufficient. Motor selection must account for starting torque, continuous heating, current, and the actual torque-speed curve.

4. BLDC Controller Selection

I recommend a programmable bidirectional FOC controller.

The controller should support:

- 6S LiPo compatibility: 25.2 V with a fully charged battery.
- CW and CCW control.
- Speed and torque/current control.
- Encoder or Hall sensor feedback.
- Motor overcurrent and overtemperature protection.
- Controlled braking and reverse commands.

A VESC has separate motor-current and battery-current parameters. At low speeds, motor phase current can be much higher than battery current. So, do not select the controller based only on an estimated battery current of 10–15 A.

5. How to Build the Freewheel System

This is an important part of your requirements.

When the cable unwinds, the drum should be able to rotate freely without being forced by the motor. However, the drum must also be prevented from spinning at an uncontrolled speed.

I recommend two operating modes:

Mode A — Winding the Cable

Motor ON

- The BLDC motor turns the drum.
- The encoder measures RPM.
- The load cell measures cable tension.
- The ESP32 controls speed and torque.

Mode B — Unwinding the Cable / Free Movement

Controlled freewheel

- Reduce motor drive torque or set it to zero.
- An electromagnetic clutch can disconnect the motor from the drum if needed.
- A low-tension brake prevents the drum from overspeeding.
- The cable tension sensor continuously monitors tension.

Recommendation: Start with a controlled freewheel system, where the motor applies a small braking torque to prevent the drum from overspeeding. If you need a fully mechanical freewheel, install an electromagnetic clutch and a separate brake.

Note: A standard one-way clutch does not allow free movement in both CW and CCW directions. Do not install one without first analyzing the mechanism.

6. Automatic Control System

Your automatic sequence:

Condition	System action

START winding	Motor runs CW
Cable tension increases	Reduce motor torque
Cable tension exceeds its limit	Stop the motor
START unwinding	Run controlled CCW or enter freewheel mode
Drum overspeeds	Apply braking
Emergency stop	Disable the drive and bring the system to a safe mechanical stop

Follow the fibre-optic cable manufacturer’s limits for pulling tension and minimum bend radius. The FOA recommends loading the cable’s strength members during pulling and monitoring tension.

7. Overall System Layout

My proposed arrangement is:

BLDC motor → electromagnetic clutch → flexible coupling → drum shaft → cable drum

Along with:

- Two bearing blocks
- Guide roller
- Tension sensor
- Encoder
- Brake
- ESP32
- Bidirectional BLDC controller

This arrangement allows the motor and drum to be disconnected for freewheel mode.

8. Three Measurements Needed Before Finalizing the Design

It would not be appropriate to buy the motor without these measurements.

1. What is the cable's actual pulling force?

Measure it with an inline digital force gauge or an S-type load cell. Record both the starting and running force while pulling the full cable over the soil.

2. What is the drum's maximum loaded radius?

The core radius is 100 mm, but the radius will increase as cable is wound onto the drum.

3. What is the cable's maximum allowable pulling tension?

Get this value from the cable's datasheet. Do not assume a generic limit for optical-fibre cable.

One important question: How much cable will you need to pull from the ground onto the drum in one operation?

- 50 metres
- 100 metres
- 200 metres
- 500 metres
- Other / not decided yet

Maximum drum radius with the cable loaded:

- Not known yet
- 150 mm
- 200 mm
- 250 mm

Once these measurements are available, we can calculate the motor’s torque, RPM, power, controller current, and battery operating time.

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